Spring 2021 Real Analysis II — Homework 3

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Denote $\mathbb{R}_+ := [0, \infty)$ and $\mathbb{R}_{++} := (0, \infty)$, $\triangle^n := \{x = (x_1, \ldots, x_n) \in \mathbb{R}^n_+ : \sum_{i=1}^m x_i = 1\}$.

1. Let X be a B* space and C a proper convex subset of X such that $0 \in \operatorname{int}(C)$. Let P be the Minkowski functional of C. Prove the following two statements: (a) $x \in \operatorname{int}(C)$ if and only if $P(x) < 1$. (b) $\overline{\operatorname{int}(C)} = \overline{C}$.

2. Let X be a Banach space. If E is a finite subset of X, then $\operatorname{conv}(E)$ is sequentially compact. [Hint: for any $\epsilon > 0$, consider a finite δ-net of $[0, 1]$ with properly chosen $\delta > 0$, and use it to construct an ϵ-net of $\operatorname{conv}(E)$.]

3. Let X be a Banach space. If E is a totally bounded subset of X, then $\operatorname{conv}(E)$ is totally bounded and $\overline{\operatorname{conv}}(E)$ is sequentially compact. [Hint: $\operatorname{conv}(N)$ is an ϵ-net of $\operatorname{conv}(E)$ if N is an ϵ-net of E.]

4. Let C be a compact convex set of a B* space X and $T : C \to C$ a continuous function. Show that T has a fixed point in C.

5. Let X be a B* space and $E \subset X$. We call $T : E \to X$ *compact* if T is continuous and $T(K)$ is sequentially compact whenever $K \subset E$ is bounded. Suppose C is a bounded closed convex set of X and $T : C \to C$ is compact. Show that T has a fixed point in C.

6. Let $A \in \mathbb{R}^{n\times n}_{++}$. Show that there exists $\lambda > 0$ and $x \in \mathbb{R}^n_+$ such that $Ax = \lambda x$. [Hint: consider $Tx = Ax/|Ax|_1$ where $|y|_1$ is the 1-norm of the vector y.]

7. Let $k : [0, 1] \times [0, 1] \to \mathbb{R}_{++}$ be continuous. Define $T : C([0, 1]) \to C([0, 1])$ by
$$(Tx)(t) = \int_0^1 k(t, s)x(s)\, \mathrm{d}s$$
for any $x \in C([0, 1])$. Show that there exists $\lambda > 0$ and a nonnegative $x \in C([0, 1])$ such that $Tx = \lambda x$.

8. Let X be a Banach space and C a bounded closed convex subset of X. Suppose $T : C \to C$ satisfies
$$\|Tx - Ty\| \le \|x - y\|$$
for all $x, y \in X$. Show that $\inf_{x\in C} \|x - Tx\| = 0$. [Hint: fix $z \in C$ and $\epsilon \in (0, 1)$, consider $T_\epsilon x := \epsilon z + (1 - \epsilon)Tx$. Show that T_ϵ has a fixed point x_ϵ in C and $\|Tx_\epsilon - x_\epsilon\| = O(\epsilon)$.]

9. Let X be a Banach space with norm $\|\cdot\|$. Show that there exists an inner product $\langle\cdot, \cdot\rangle : X \times X \to \mathbb{R}$ such that $\langle x, x\rangle = \|x\|^2$ if and only if $\|\cdot\|$ satisfies the parallelogram law:
$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad \forall\, x, y \in X.$$
[Hint: first show that $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$ and $\langle -x, y\rangle = \langle x, -y\rangle = -\langle x, y\rangle$ for all $x, y, z \in X$. Then show that $\langle rx, y\rangle = r\langle x, y\rangle$ for all $r \in \mathbb{R}$. Finally show that $|\langle \lambda x, y\rangle - r_k\langle x, y\rangle| \to 0$ as $r_k \to \lambda$.]

10. Show that there is no inner product that can induce the norm $\|x\| = \max_{0\le t\le 1} |x(t)|$ on $C([0, 1])$.